

Section I

10 marks

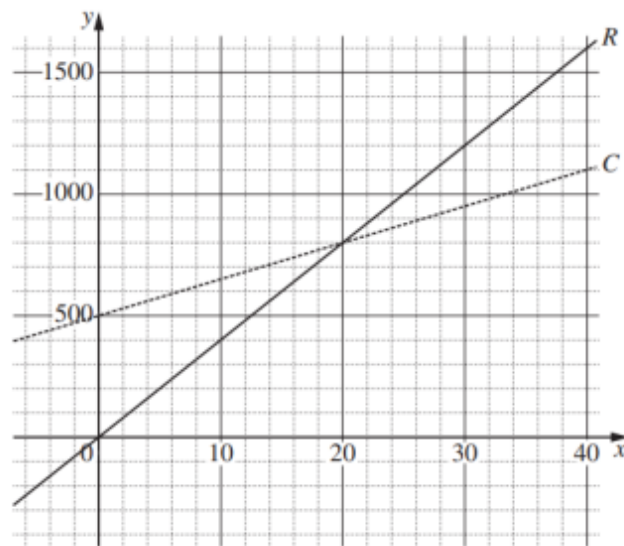
Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

- 1 A small business makes and sells bird houses.

Technology was used to draw straight-line graphs to represent the cost of making bird houses (C) and the revenue from selling bird houses (R). The x – axis displays the number of bird houses and the y – axis displays the cost/revenue in dollars.



How many bird houses need to be sold to break even?

- A. 10
B. 20
C. 30
D. 40
- 2 The relationship between number of beers consumed (x) and blood alcohol content (y) was studied in 18 male college students by using least squares regression. The following regression equation was obtained from this study $y = -0.0127 + 0.0180x$.

With regards to the study and the regression equation, which of the following statements is correct?

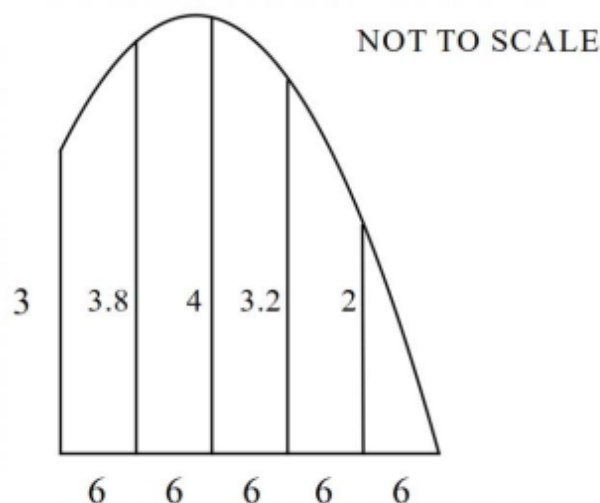
- A. Each beer consumed increases blood alcohol content by 1.27%.
B. On average it takes 1.8 beers to increase blood alcohol content by 1%.
C. Each beer consumed increases blood alcohol content by 0.0180.
D. Each beer consumed decreases blood alcohol content by 0.0127.

- 3 The table below shows the future value of a \$1 annuity for varying interest rates over different time periods.

Time period	Interest rate				
	1%	2%	3%	4%	5%
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.526
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

What is the present value of an annuity, correct to the nearest dollar, that would provide a future value of \$62 000 after 3 years at 2% per annum, compounded half-yearly?

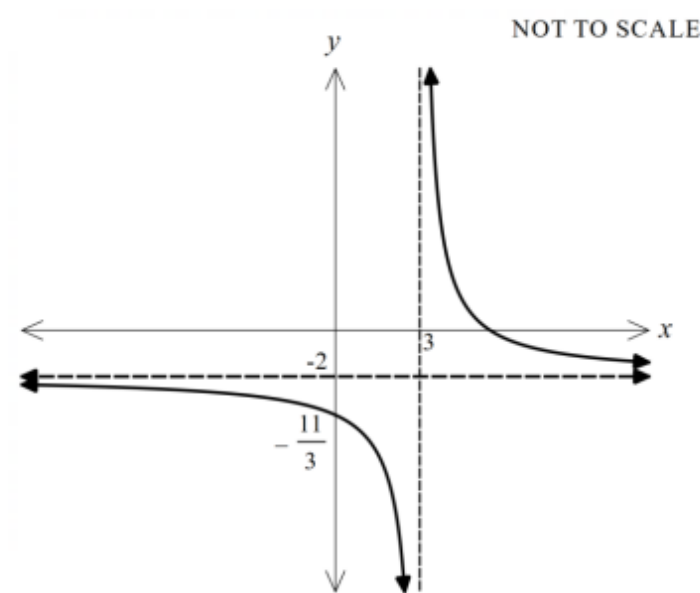
- A. \$9, 828.63
 B. \$10, 078.02
 C. \$20, 258.79
 D. \$20, 461.37
- 4 The diagram below shows a poppy field. All measurements are in metres.



What is an approximate value for the area of the poppy field using the trapezoidal rule with 5 sub-intervals?

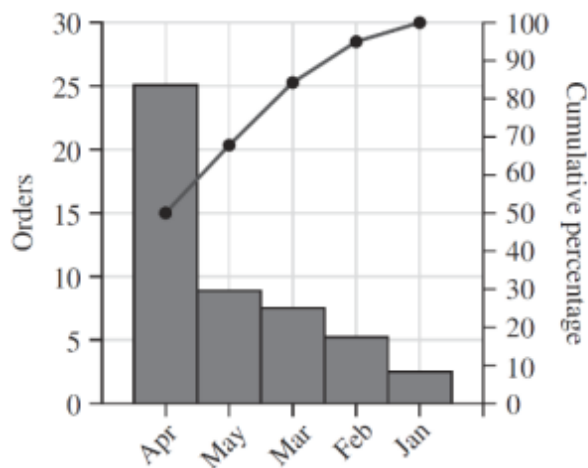
- A. 67.5m^2
 B. 72.5m^2
 C. 81m^2
 D. 87m^2

- 5 Consider the graph of $y = \frac{k}{x+b} + c$ below, with y -intercept at $\left(0, -\frac{11}{3}\right)$.



Which of the following statements is true?

- A. $k < b < c$
 - B. $b < k < c$
 - C. $k < c < b$
 - D. $b < c < k$
- 6 The Pareto chart below shows the orders received by a business for five months.



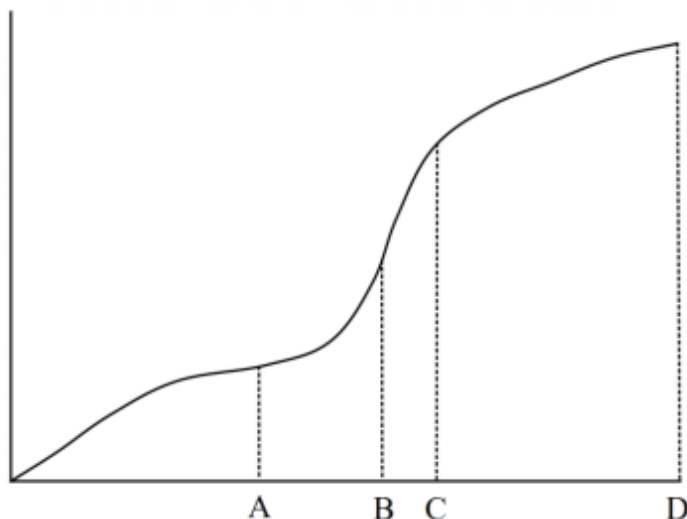
What percentage of orders were received in May?

- A. 18%
- B. 30%
- C. 45%
- D. 69%

7 Let $g(x) = 1 - f(x)$. If $\int_1^4 f(x) dx = 7$, what is the value of $\int_1^4 g(x) dx$?

- A. -6
- B. -4
- C. 8
- D. 10

8 The cumulative distribution function for the continuous random variable X is graphed below.



What is the mode of the probability density function?

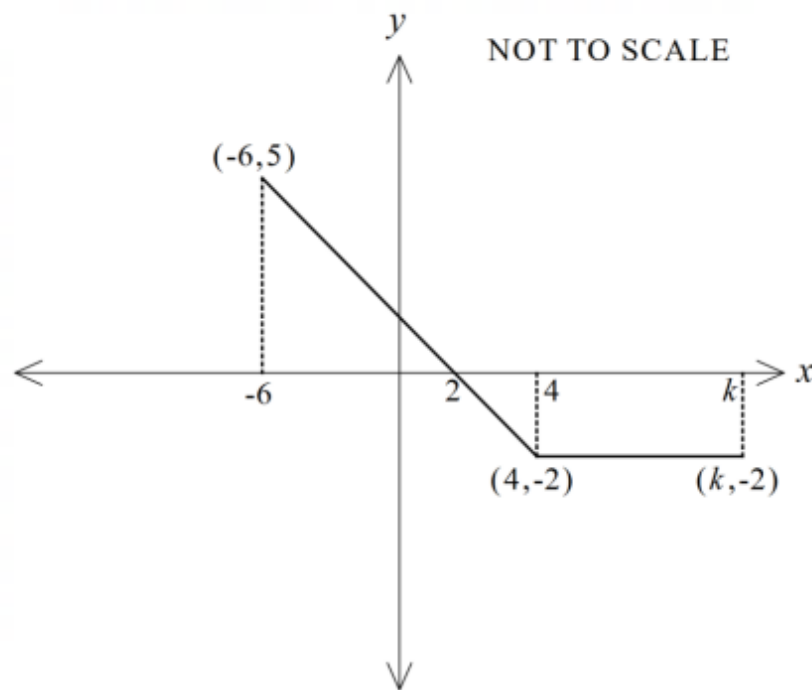
- A. A
- B. B
- C. C
- D. D

9 It is known that $f(x)$ is an odd function and $g(x)$ is an even function.

Given that $f(2) = 2$ and $g(2) = -2$, what is the value of $f(g(-2)) + g(f(-2))$?

- A. -4
- B. -2
- C. 0
- D. 4

- 10 By using the graph below, which value of k satisfies $\int_{-6}^k f(x) dx = 0$?



- A. 6
- B. 11
- C. 12
- D. 13

End of Section I

Section II

90 marks

Attempt Questions 11-32

Allow about 2 hours and 45 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response. Your responses should include relevant mathematical reasoning.

Question 11 (2 marks)

Simplify $\frac{2}{x-1} - \frac{6}{(x-1)(2x+1)}$. 2

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Question 12 (3 marks)

Solve the equation $2\sec \theta - 1 = 9$, $0 \leq \theta \leq 2\pi$. Give your answer correct to two decimal places. 3

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Question 13 (5 marks)

The probability distribution of a discrete random variable X is shown in the table below, where a , b and c are constants.

x	1	2	3	4
$P(X = x)$	a	b	b	c

The cumulative distribution function of X is shown in the table below where d and e are constants.

x	1	2	3	4
$F(x)$	$\frac{1}{6}$	d	$\frac{2}{3}$	e

- a) Determine the value of each of the constants a, b, c, d and e .

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- b) The discrete random variable Y is defined as $Y = 10 - 3X$.

Find the value of $P(Y > X)$.

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Question 14 (7 marks)

Differentiate each of the following:

a) $y = x^4 + 2x^{\frac{5}{2}}$. **1**

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b) $y = e^{2x} \tan x$. **2**

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c) $y = (x^3 - 3 \ln x)^5$. **2**

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d) $y = \frac{4x}{\cos 3x}$. **2**

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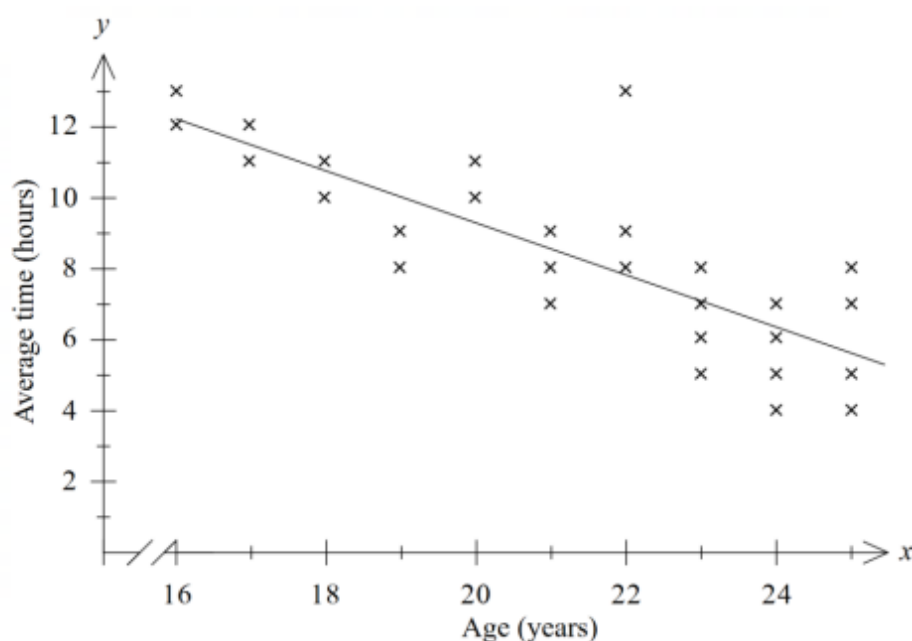
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Question 15 (4 marks)

In a training scheme for young people, the average time taken for each age group to reach a certain level of proficiency was measured.

After collecting the data, it is found that the correlation coefficient is -0.804 .

A scatterplot showing the data is drawn. The line of best fit with equation $y = 23.96 - 0.73x$, is also drawn.



Describe and interpret the data and other information provided, with reference to the context given. 4

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Question 16 (3 marks)

Seats in a theatre are arranged in rows. The number of seats in this theatre form the terms of an arithmetic series.

The sixth row has 23 seats, and the fifteenth row has 50 seats. If the theatre has 20 rows, how many seats are in the theatre?

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Question 17 (2 marks)

Prove $\frac{1}{\sec x - \tan x} + \frac{1}{\sec x + \tan x} = 2 \sec x$.

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Question 18 (5 marks)

During the winter, Andrew attends weekly business meetings in Queenstown and always travels to these meetings by car.

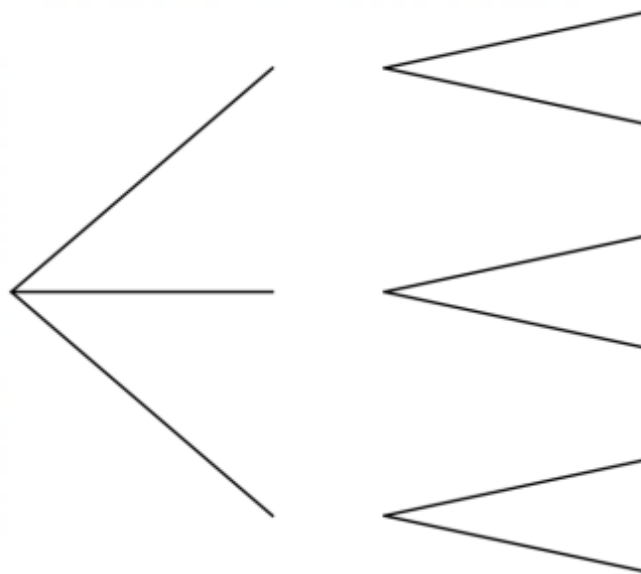
The probability of being dry, raining or snowing during his travel to these meetings is

$\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{6}$, respectively. The respective probabilities of Andrew arriving on time when it is

dry, raining or snowing are $\frac{4}{5}$, $\frac{2}{5}$ and $\frac{1}{10}$.

- a) Complete the probability tree diagram below, showing all outcomes.

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- b) Determine the probability that Andrew will arrive late to his next winter business meeting.

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- c) Andrew arrived late for his meeting last week. Find the probability that it was raining on that day. 2

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Question 19 (4 marks)

A random variable is normally distributed with a mean of 0 and a standard deviation of 1. The table provided on page 33 gives the probability that this random variable lies below z for some positive values of z .

The weekly mileages covered by a sales rep, M miles, are normally distributed with a mean of 365 and variance of 600.

Find the probability that in a random week the rep will cover a mileage between 380 and 410. 4

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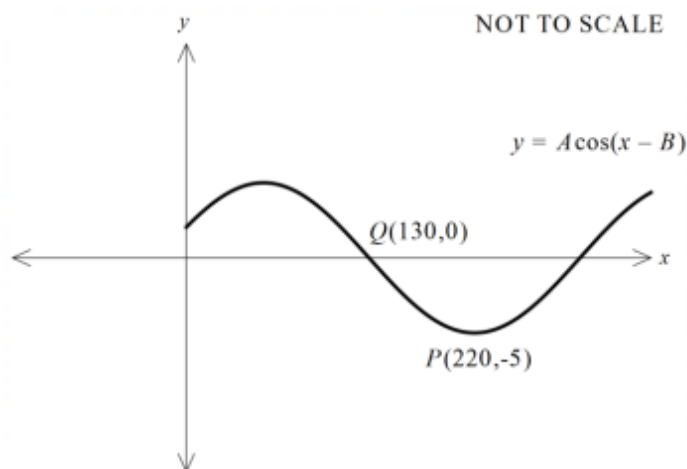
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Question 20 (2 marks)

The figure below shows the graph of the curve with equation $y = A\cos(x - B)$,

$0^\circ \leq x \leq 360^\circ$, where A and B are positive constants with $0^\circ < B < 90^\circ$.



The graph meets the x axis at point $Q(130, 0)$ and the point $P(220, -5)$ is the minimum point of the curve. State the value of A and the value of B .

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Question 21 (6 marks)

Find the following integrals:

a) $\int \sqrt{x} - 4 \cos 2x \, dx$ **2**

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b) $\int 3^{2x+1} \, dx$ **2**

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c) $\int \frac{3}{x} + \frac{4}{x^2} - \frac{2}{x^3} \, dx$ **2**

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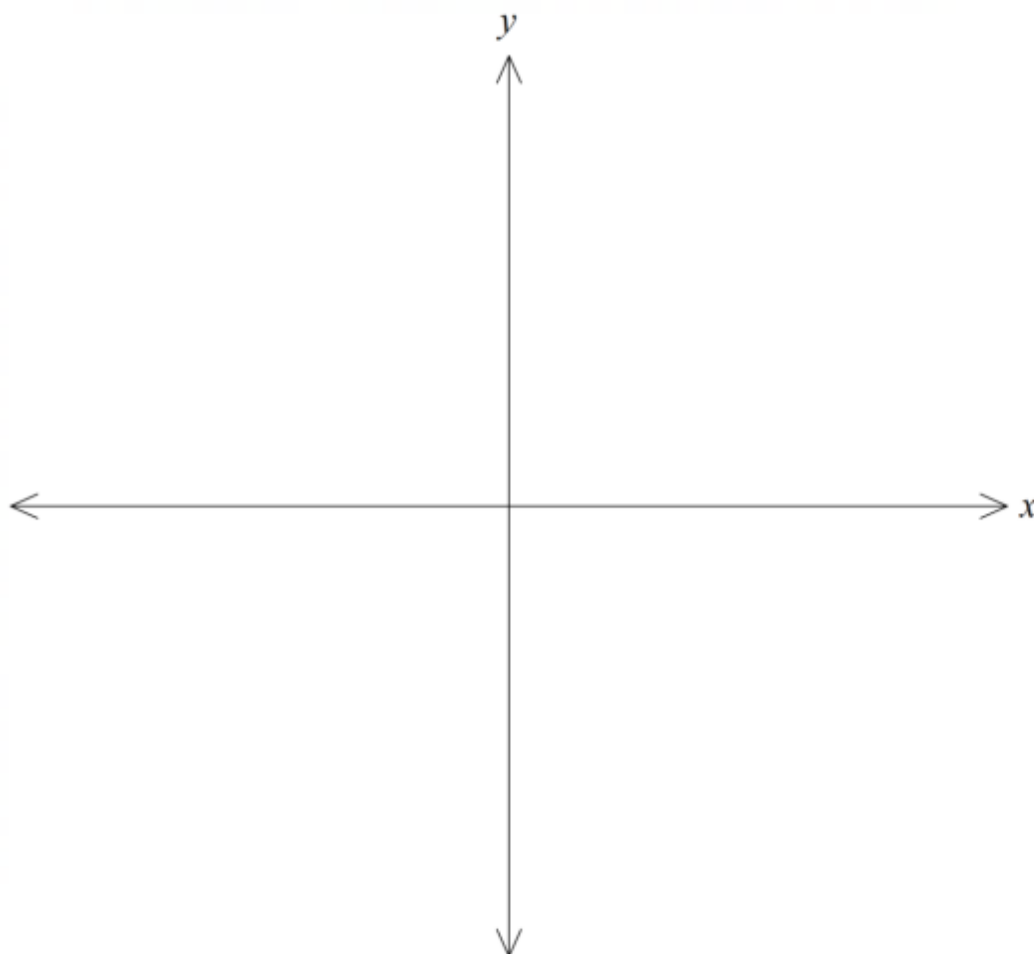
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Question 22 (4 marks)

- a) Sketch the graphs of the functions $f(x) = x - 2$ and $g(x) = (2x - 1)(2 - x)$ showing the intercepts.

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- b) Hence, or otherwise, solve the inequality $x - 2 > (2x - 1)(2 - x)$.

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Question 23 (4 marks)

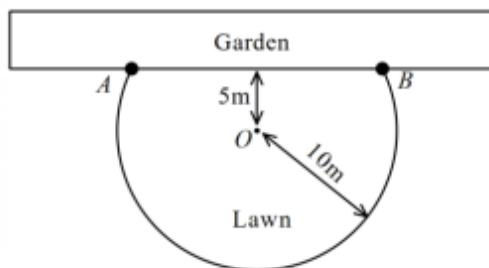
Find the global maximum and minimum values of $y = x^3 - 3x^2 + 5$, where $-\frac{1}{2} \leq x \leq 4$.

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Question 24 (6 marks)

A water sprinkler covers a circular lawn area of radius 10 metres. The sprinkler (O) is placed 5 metres from a rectangular garden bed. Garden stakes are placed at A and B , as shown in the diagram below.

NOT TO SCALE



- a) Show that $\angle AOB = \frac{2\pi}{3}$.

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- b) Show that the major arc length of the lawn area covered by the sprinkler is $\frac{40\pi}{3}$ m.

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- c) Find the exact area of the lawn that the sprinkler will cover.

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This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines, providing a guide for letter height and placement. The lines are evenly spaced across the entire page, leaving ample room for writing practice. There is no text or other markings on the page.

Question 25 (3 marks)

a) Differentiate $y = \frac{\ln x}{x}$.

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b) Hence, evaluate $\int_e^{e^2} \frac{2 - \ln x^2}{x^2} dx$.

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Question 26 (4 marks)

A particle P is moving on the x axis and its velocity $v \text{ ms}^{-1}$, t seconds after a given instant, is given by $v = t^2(3 - t)$, $t \geq 0$.

When $t = 2$, P is observed to be 4m from the origin O , in the positive direction of the x - axis.

- a) Find the acceleration of P when $t = 2$.

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- b) The particle is initially at rest and is also at rest when $t = T$. Determine the distance of P from O when $t = T$.

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Question 27 (5 marks)

A car dealership has a car for sale for a cash price of \$20 000. It can also be bought on terms over three years. The first six months are interest free. After the first six months, reducible interest is charged at the rate of 1% per month, calculated at the end of each month.

Repayments are to be made in equal monthly instalments beginning at the end of the first month.

A customer buys the car on these terms and agrees to monthly repayments of \$ M . Let \$ A_n be the amount owing at the end of the n^{th} month.

- a) Find an expression for A_6 . **1**

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- b) Show that $A_8 = (20\,000 - 6M)1.01^2 - M(1 + 1.01)$. **1**

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c) Find the value of M .

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[illegible]

Question 28(8 marks)

Consider $f(x) = e^{2x+1} x^4$.

- a) Find the stationary point(s) and determine their nature.

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[illegible]

b) Find the point(s) of inflection.

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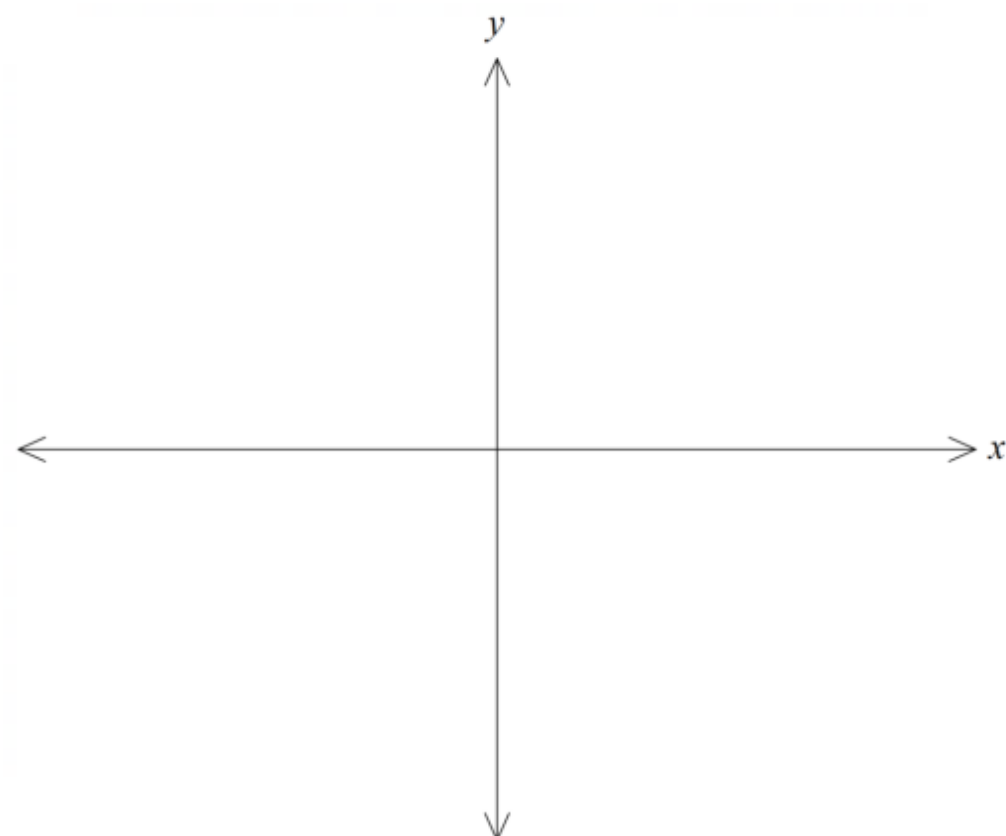
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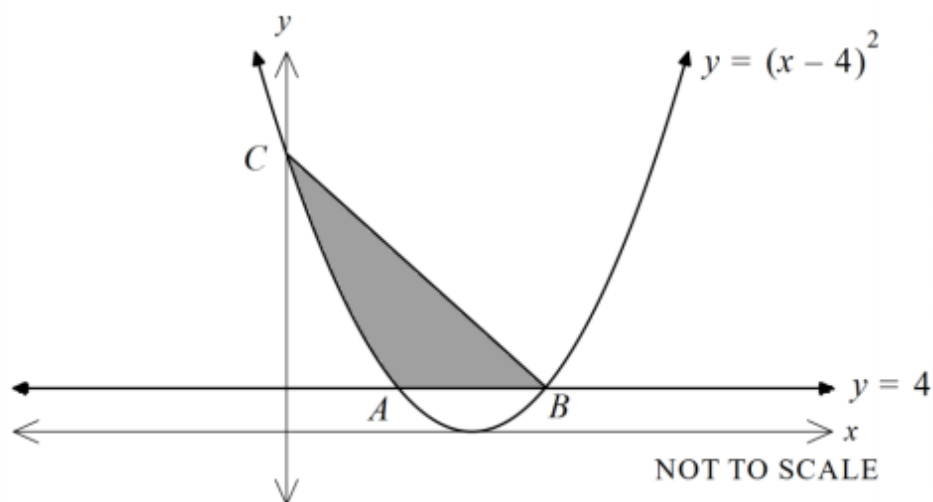
c) Sketch the graph of $f(x)$ showing all features.

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Question 29 (3 marks)

The diagram below shows the curve with equation $y = (x - 4)^2$, intersected by the straight line with equation $y = 4$, at the points A and B . The curve meets the y axis at the point C .



Calculate the exact area of the shaded region, bounded by the curve and the straight-line segments AB and BC .

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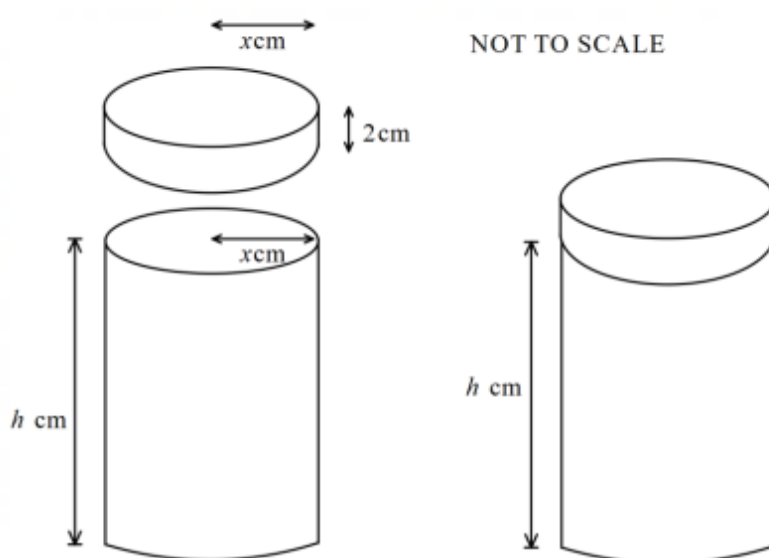
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Question 30 (5 marks)

The figure below shows the design of coffee jar with a lid sitting directly on top.



The jar is in the shape of a right circular cylinder of radius $x \text{ cm}$. It is fitted with a lid of height 2 cm , which fits tightly on the top of the jar, so it may be assumed that it has the same radius as the jar. The jar and its lid are made of thin sheet metal and there is no wastage.

The total metal used to make the jar and its lid is $190\pi \text{ cm}^2$.

- a) Show that the volume of the jar, $V \text{ cm}^3$, is given by $V = \pi(95x - 2x^2 - x^3)$.

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- b) Find the maximum volume of the jar, leaving your answer in exact form.

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[illegible]

Question 31 (3 marks)

Given $f(x) = \frac{x+1}{2x-1}$ and $f(g(x)) = \frac{3x+2}{3x-5}$, find $g(x)$.

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This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines, providing a guide for letter height and placement. The lines are evenly spaced across the entire page, which is otherwise blank.

Question 32 (2 marks)

Find $\int \frac{1}{\sqrt{x^{\frac{3}{2}} + 5x}} dx.$

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[illegible]

End of Exam

Standard Normal Probabilities

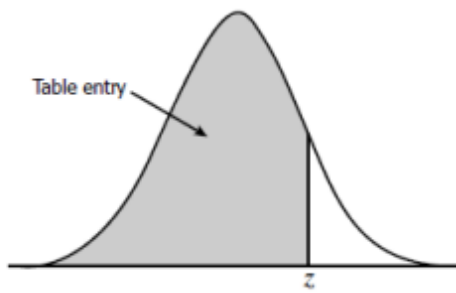


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998